DSA 8070 R Session 12: Cluster Analysis

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K-Means Clustering

K-means partitions observations into K clusters by minimizing the total within-cluster sum of squares. Each observation is assigned to the cluster with the nearest centroid.

Simulated Example

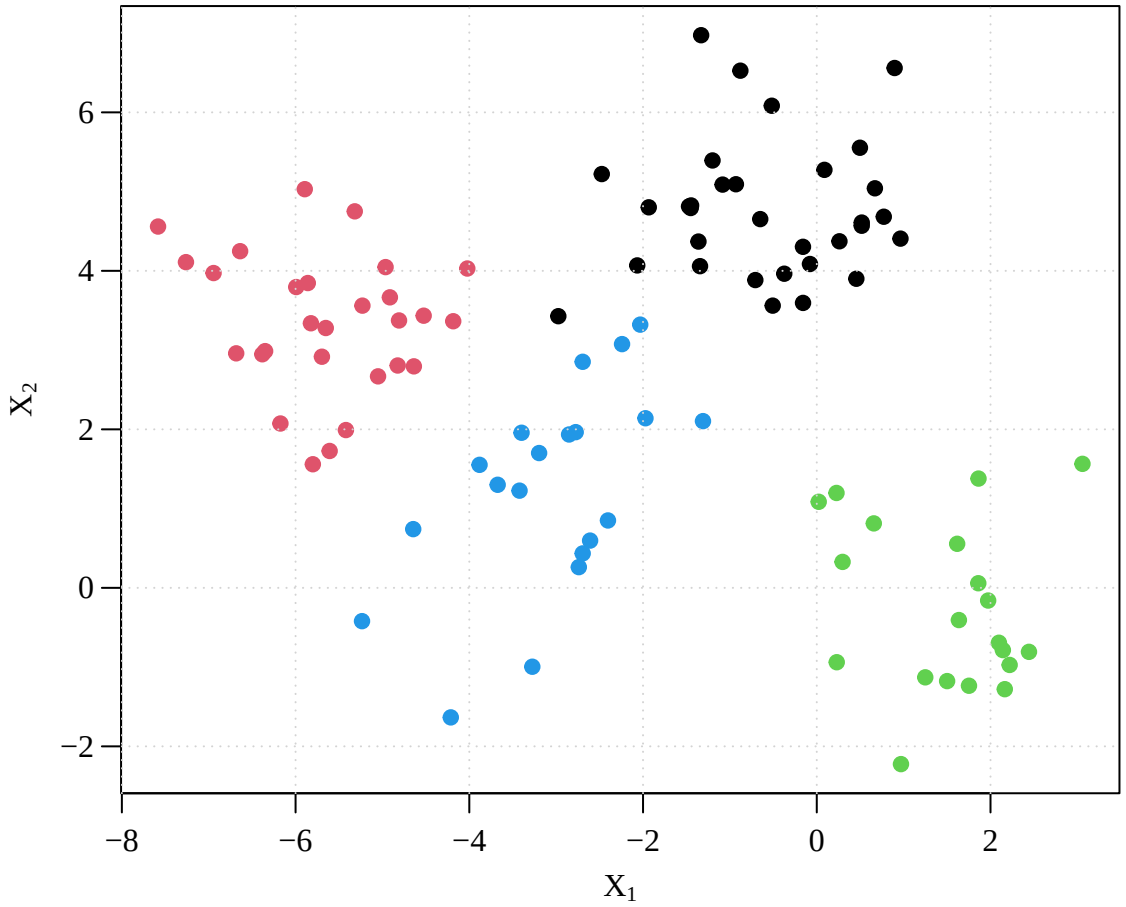
We first simulate four groups in two dimensions and examine how well K-means recovers the underlying cluster structure.

```
set.seed(101)

# Generate observations around four randomly selected centers
x <- matrix(rnorm(100 * 2), 100, 2)
xmean <- matrix(rnorm(8, sd = 4), 4, 2)

which <- sample(1:4, 100, replace = TRUE)
x <- x + xmean[which, ]

# Plot the true group memberships
par(mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0), las = 1, family = "serif")
plot(x, col = which, pch = 19, xlab = expression(X[1]), ylab = expression(X[2]))
grid()
```

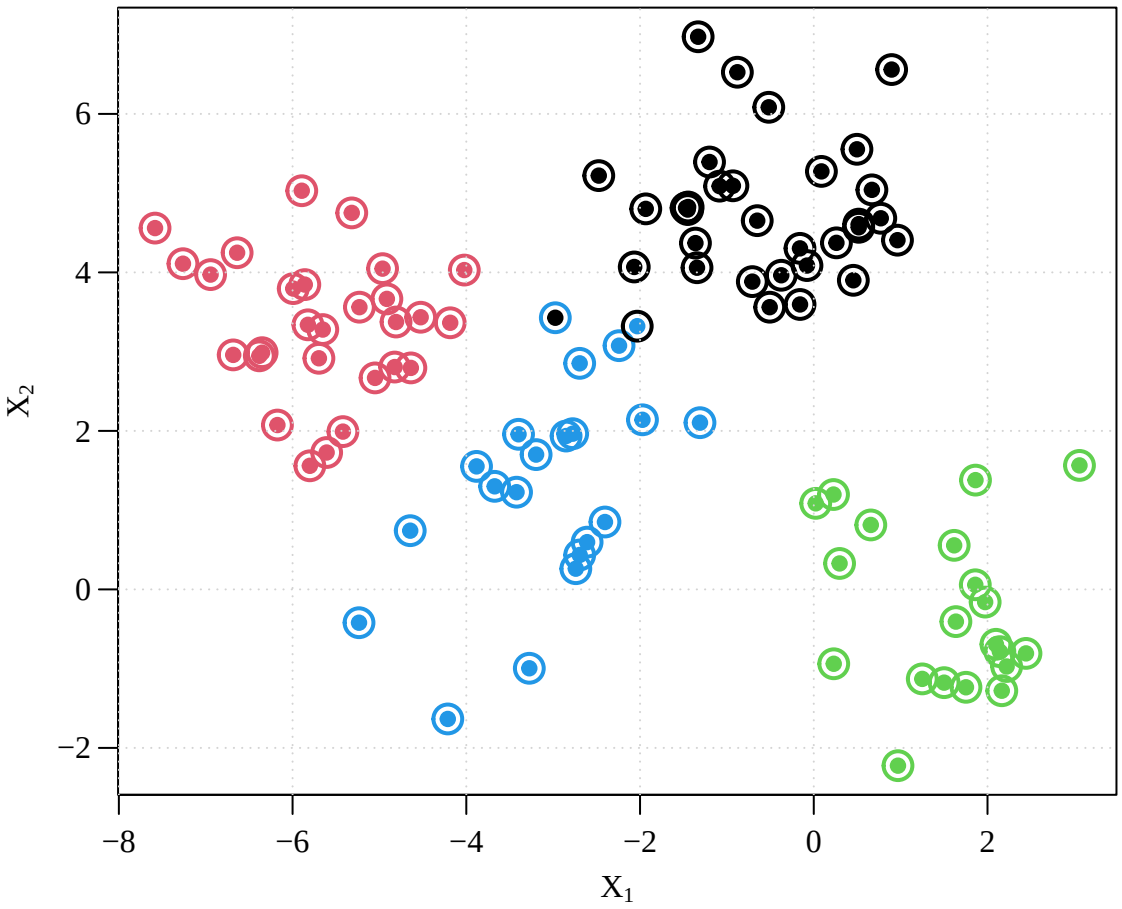



```r
# Fit K-means with K = 4 using multiple random starts
km.out <- kmeans(x, centers = 4, nstart = 15)
km.out
```

```
## K-means clustering with 4 clusters of sizes 32, 28, 20, 20
##
## Cluster means:
##         [,1]       [,2]
## 1 -0.5787702  4.7639233
## 2 -5.6518323  3.3513316
## 3  1.4989983 -0.2412154
## 4 -3.1104142  1.2535711
##
## Clustering vector:
##   [1] 2 4 1 2 4 1 2 4 1 1 3 1 1 3 4 3 2 3 2 2 2 2 2 3 1 1 4 2 4 1 2 3 2 4 4 3 3
##  [38] 4 3 3 2 4 4 2 2 3 2 1 2 4 2 1 1 3 3 4 3 1 1 1 4 2 2 2 4 4 1 1 3 2 2 1 1 3
##  [75] 1 3 2 1 1 1 4 1 4 1 2 3 1 2 2 1 1 4 2 4 1 1 3 3 1 1
##
## Within cluster sum of squares by cluster:
## [1] 53.04203 42.40322 34.95921 48.52107
##  (between_SS / total_SS =  85.7 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
```

```
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```r
# Compare estimated clusters with the true groups
par(mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0), las = 1, family = "serif")
plot(x, col = km.out$cluster, cex = 2, pch = 1, lwd = 2,
     xlab = expression(X[1]), ylab = expression(X[2]))
points(x, col = which, pch = 19)
grid()
```



The open circles represent the estimated K-means memberships, while the filled points represent the true groups. Note that cluster labels are arbitrary: two clusterings can be identical even if their numerical labels differ.

Geyser Example

The `faithful` dataset contains eruption durations and waiting times for the Old Faithful geyser. We compare K-means solutions with different numbers of clusters to see how the partition changes as K increases.

```r
set.seed(101)
km2.faithful <- kmeans(faithful, 2, nstart = 25)
km3.faithful <- kmeans(faithful, 3, nstart = 25)
km4.faithful <- kmeans(faithful, 4, nstart = 25)

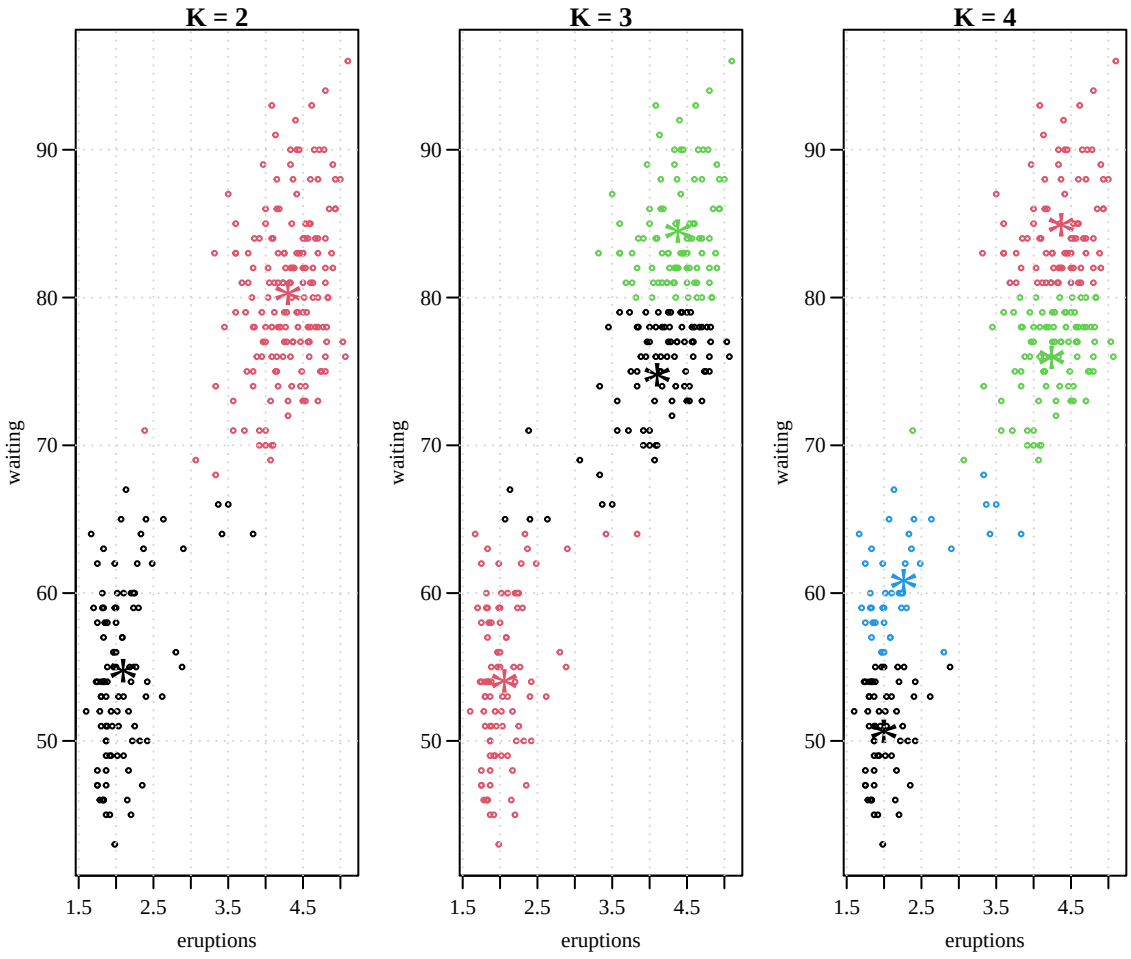
# Compare cluster assignments and centroids
```

```r
par(las = 1, mfrow = c(1, 3), mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0),
    family = "serif")

plot(faithful, col = km2.faithful$cluster, cex = 0.5, main = "K = 2")
points(km2.faithful$centers, cex = 3, pch = "*", col = 1:2)
grid()

plot(faithful, col = km3.faithful$cluster, cex = 0.5, main = "K = 3")
points(km3.faithful$centers, cex = 3, pch = "*", col = 1:3)
grid()

plot(faithful, col = km4.faithful$cluster, cex = 0.5, main = "K = 4")
points(km4.faithful$centers, cex = 3, pch = "*", col = 1:4)
grid()
```




```r
par(mfrow = c(1, 1))
```

Increasing K reduces the within-cluster sum of squares, but it does not necessarily produce a more meaningful clustering. In this example, consider whether the additional clusters reveal distinct groups or simply subdivide existing ones.

US State Facts and Figures Example

We now apply K-means to four socioeconomic variables from the `state.x77` dataset. Because these variables have different measurement scales, we first examine the results using the raw data.

```
# Select four state-level variables
vars <- c("Income", "Illiteracy", "Life Exp", "HS Grad")
head(state.x77[, vars])
```

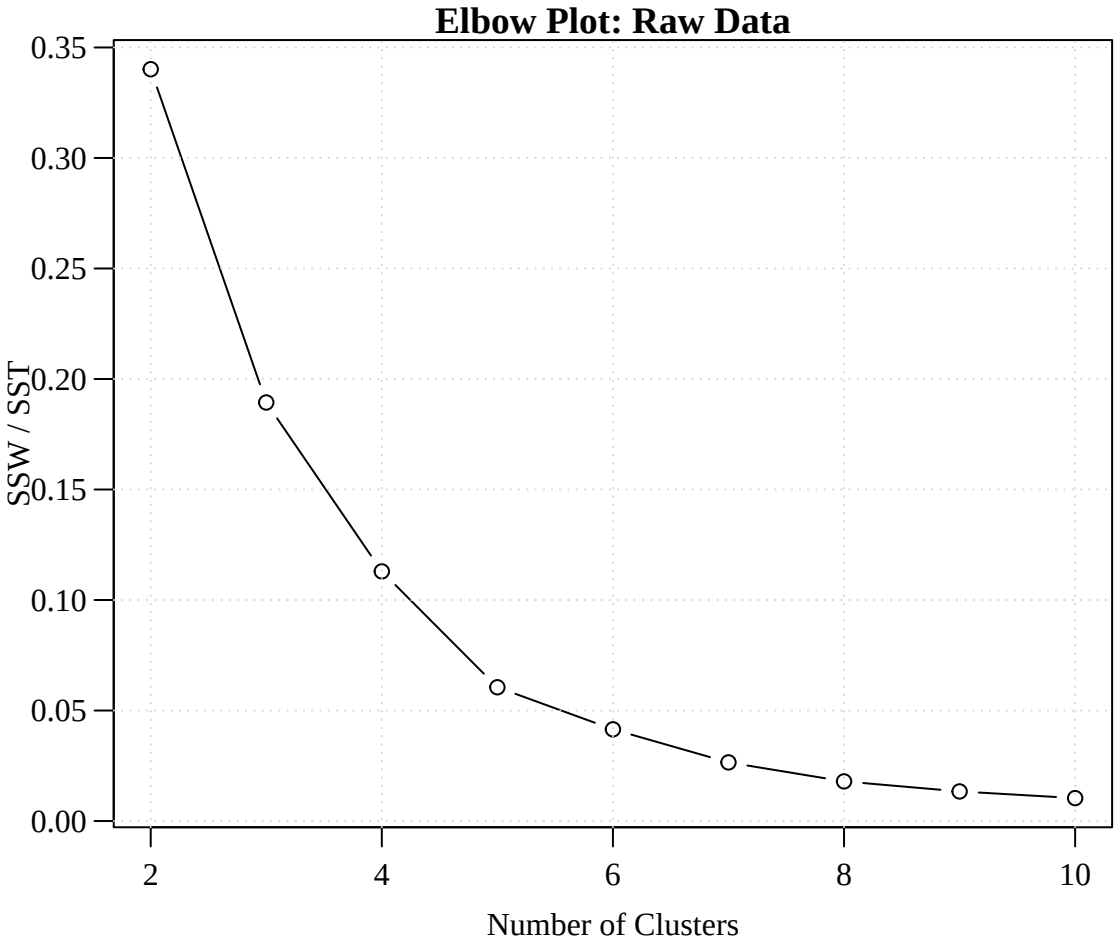
```
##            Income Illiteracy Life Exp HS Grad
## Alabama      3624        2.1    69.05    41.3
## Alaska       6315        1.5    69.31    66.7
## Arizona      4530        1.8    70.55    58.1
## Arkansas     3378        1.9    70.66    39.9
## California   5114        1.1    71.71    62.6
## Colorado     4884        0.7    72.06    63.9
```

```
# Fit K-means for K = 2, ..., 10 using raw data
kmlist <- vector("list", 9)

for (k in 2:10) {
  set.seed(1)
  kmlist[[k - 1]] <- kmeans(state.x77[, vars], k, nstart = 100)
}

# Examine the proportion of unexplained variation
tot.withinss <- sapply(kmlist, function(x) x$tot.withinss)

par(mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0), las = 1, family = "serif")
plot(2:10, tot.withinss / kmlist[[1]]$totss, type = "b",
     xlab = "Number of Clusters",
     ylab = "SSW / SST",
     main = "Elbow Plot: Raw Data")
grid()
```



The ratio SSW/SST measures the proportion of total variation remaining within clusters. A noticeable change in the rate of decrease may suggest a reasonable choice of K, although the elbow is not always clearly defined.

We next visualize the cluster assignments on a US map.

```
library(maps)
library(RColorBrewer)

par(mfrow = c(2, 2), mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0),
    las = 1, family = "serif")

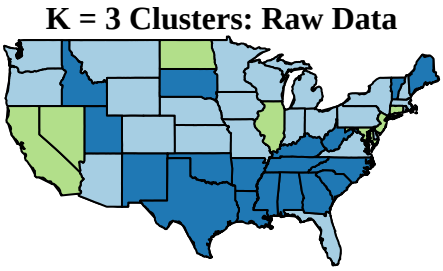
for (k in 3:6) {
  maps::map(database = "state")
  title(paste0("K = ", k, " Clusters: Raw Data"))

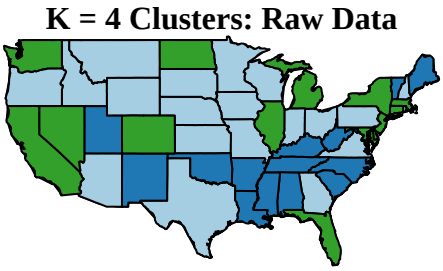
  cols <- brewer.pal(k, "Paired")

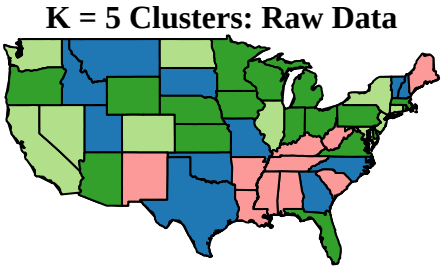
  for (j in 1:k) {
    ix <- names(which(kmlist[[k - 1]]$cluster == j))

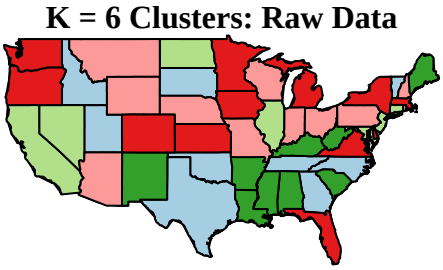
    if (length(ix) > 1) {
      maps::map(database = "state", regions = ix,
                col = cols[j], fill = TRUE, add = TRUE)
    }
```

```
  }
}
```










```
par(mfrow = c(1, 1))
```

Important consideration: K-means uses Euclidean distances, so variables with larger numerical scales can dominate the clustering. The state-level results based on raw data may therefore be driven disproportionately by variables such as income.

A useful extension is to repeat the analysis using standardized variables and compare the resulting partitions.

Hierarchical Clustering

Unlike K-means, hierarchical clustering builds a nested sequence of clusters without requiring us to specify K in advance. The resulting dendrogram illustrates how observations or groups are merged.

US State Facts and Figures Example

We use the same four state-level variables to compare different linkage methods.

```
# Examine variable scales
apply(state.x77[, vars], 2, mean)
```

```
##     Income Illiteracy   Life Exp    HS Grad
##  4435.8000     1.1700    70.8786    53.1080
```

```r
apply(state.x77[, vars], 2, sd)
```

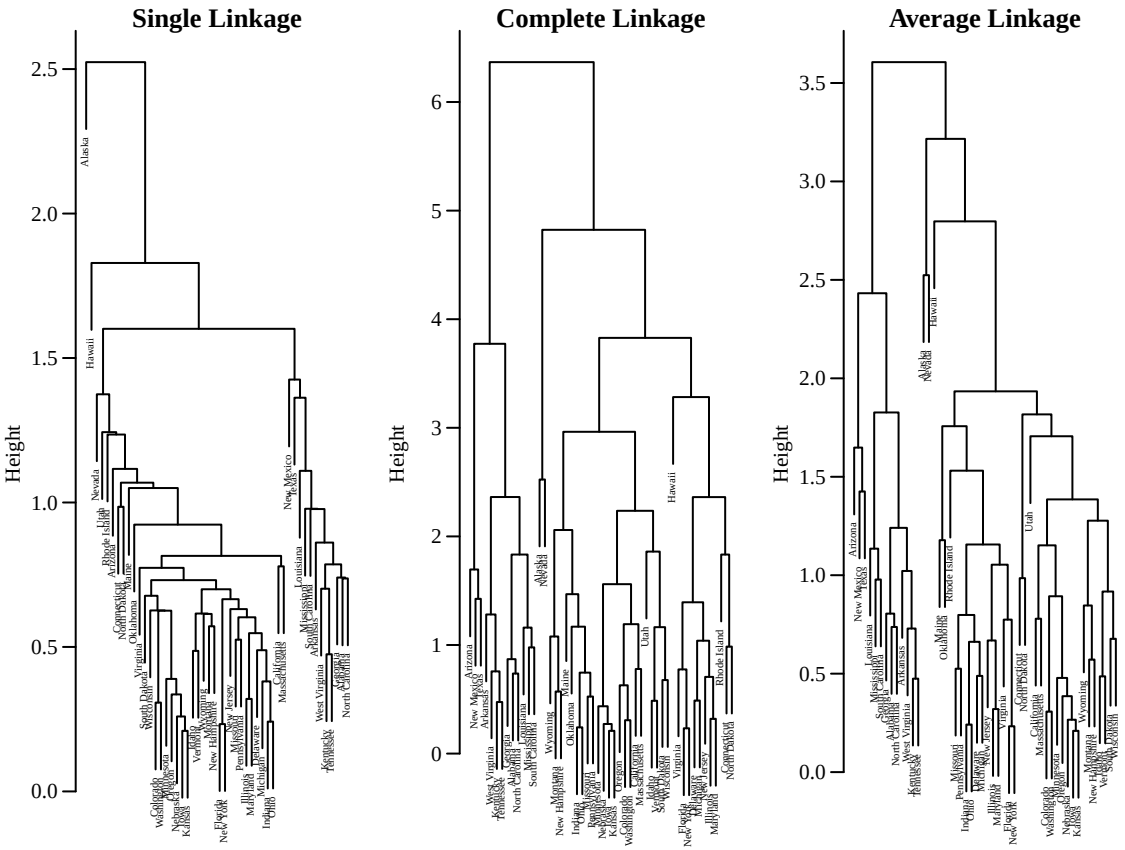
```
##      Income  Illiteracy    Life Exp     HS Grad
## 614.4699392   0.6095331   1.3423936   8.0769978
```

```r
# Compute Euclidean distances before and after standardization
distraw <- dist(state.x77[, vars])
diststd <- dist(scale(state.x77[, vars]))

# Apply three linkage methods to standardized data
hcstdSL <- hclust(diststd, method = "single")
hcstdCL <- hclust(diststd, method = "complete")
hcstdAL <- hclust(diststd, method = "average")

# Compare dendrograms
par(las = 1, mfrow = c(1, 3), mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0),
    family = "serif")

plot(hcstdSL, cex = 0.5, main = "Single Linkage", xlab = "", sub = "")
plot(hcstdCL, cex = 0.5, main = "Complete Linkage", xlab = "", sub = "")
plot(hcstdAL, cex = 0.5, main = "Average Linkage", xlab = "", sub = "")
```




```r
par(mfrow = c(1, 1))
```

Although all three methods use the same standardized observations and Euclidean distances, they can produce substantially different dendrograms.

Single linkage is sensitive to chaining, complete linkage tends to favor compact clusters, and average linkage provides an intermediate approach. The height of each merge represents the dissimilarity between the clusters being combined, as defined by the chosen linkage method.

Model-based

Model-based clustering assumes that the observations arise from a mixture of probability distributions, commonly multivariate Gaussian distributions. Unlike K-means, it allows clusters to have different covariance structures and provides probabilities of cluster membership.

Geyser Example

We return to the Old Faithful geyser data and fit Gaussian mixture models. The `mclust` package uses BIC to compare candidate models with different numbers of components and covariance structures.

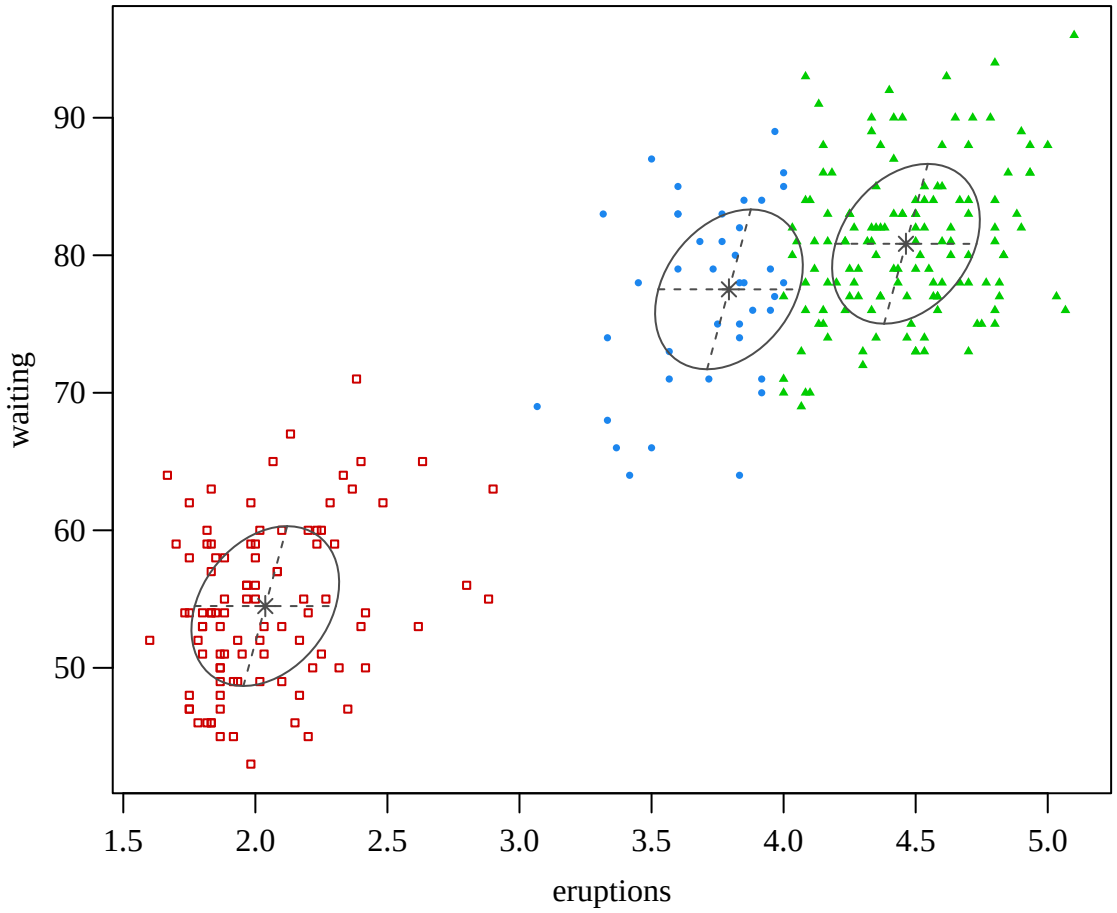
```r
library(mclust)

# Compare Gaussian mixture models using BIC
BIC <- mclustBIC(faithful)
model1 <- Mclust(faithful, x = BIC)

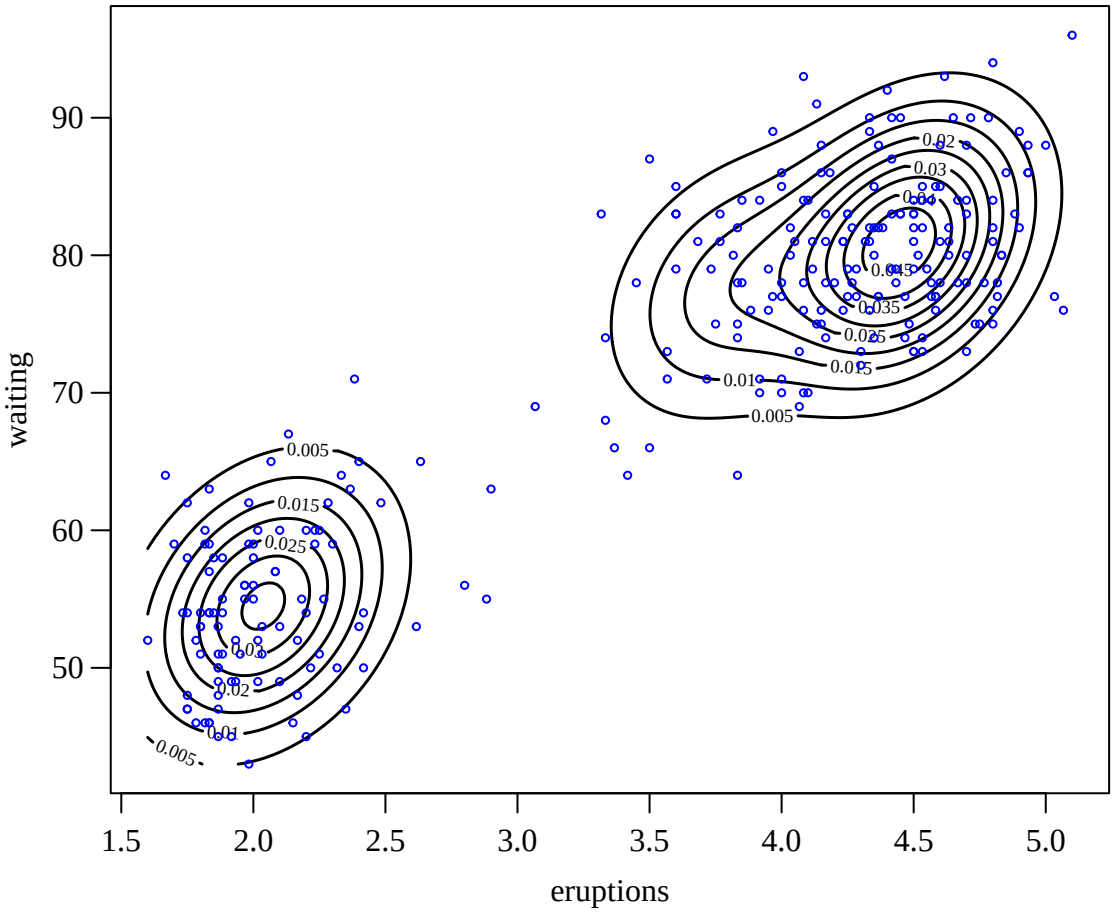
# Inspect the selected model
summary(model1)
```

```
## ----------------------------------------------------
## Gaussian finite mixture model fitted by EM algorithm
## ----------------------------------------------------
##
## Mclust EEE (ellipsoidal, equal volume, shape and orientation) model with 3
## components:
##
##  log-likelihood   n df       BIC       ICL
##       -1126.326 272 11 -2314.316 -2357.824
##
## Clustering table:
##   1   2   3
##  40  97 135
```

```r
# Visualize the estimated classification
par(las = 1, mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0), family = "serif")
plot(model1, what = "classification", cex = 0.5, las = 1)
```




```
# Visualize the fitted mixture density
par(las = 1, mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0), family = "serif")
plot(model1, what = "density", col = "black", lwd = 1.5, las = 1)
points(faithful, col = "blue", cex = 0.5)
```




```
# Bootstrap likelihood ratio tests for the number of components
set.seed(101)
LRT <- mclustBootstrapLRT(faithful, modelName = "VVV")
LRT
```

```
## ----------------------------------------------------------
## Bootstrap sequential LRT for the number of mixture components
## ----------------------------------------------------------
## Model        = VVV
## Replications = 999
##                LRTS bootstrap p-value
## 1 vs 2   319.065354             0.001
## 2 vs 3     6.130516             0.551
```

The classification plot shows the most likely component for each observation, while the density plot illustrates the fitted Gaussian mixture distribution.

Unlike K-means, model-based clustering can represent elliptical clusters with different orientations and spreads. The selected number of components depends on the assumed model family and the BIC criterion.

The bootstrap likelihood ratio test provides another approach to comparing numbers of mixture components under a specified covariance model. Its conclusions need not agree with the BIC-selected model.

Fisher's Iris Data Example

The Iris dataset contains four measurements for 150 flowers belonging to three known species. We fit a Gaussian mixture model **without using the species labels** and then examine the resulting clusters.

```
data(iris)

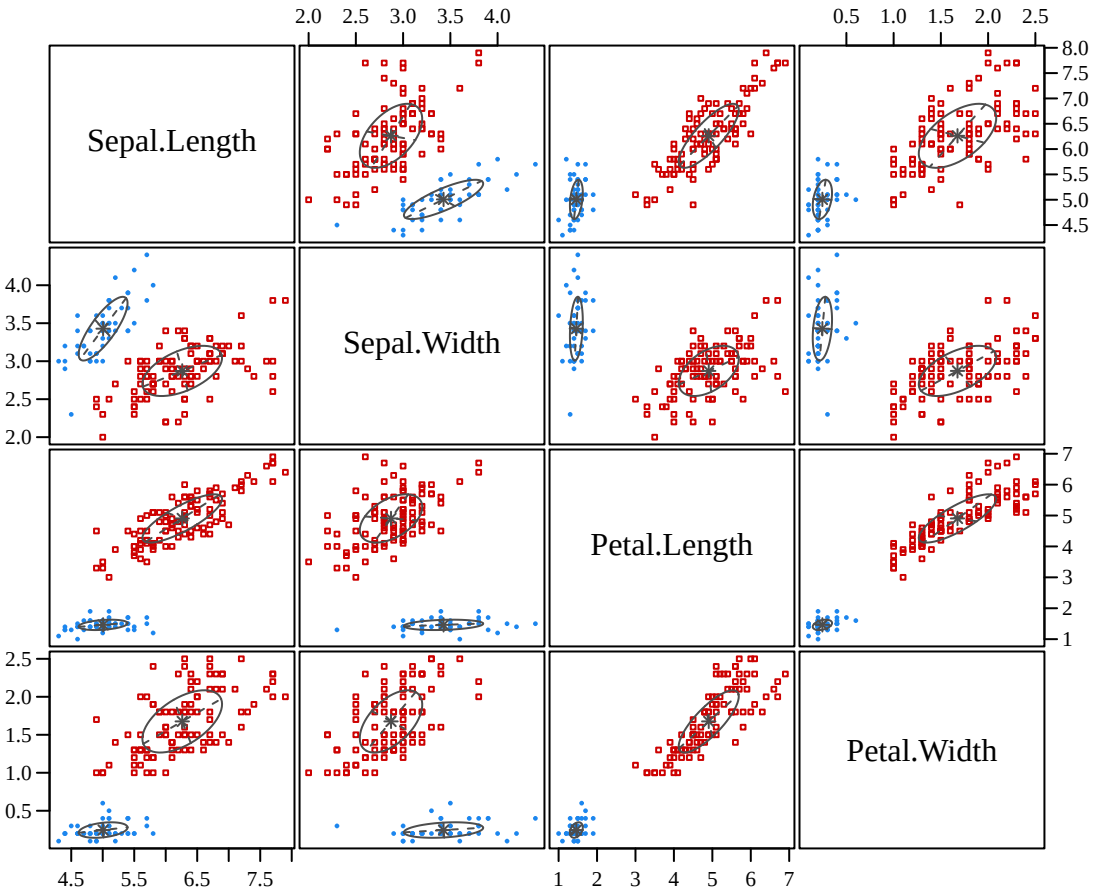
# Use flower measurements only; exclude known species labels
dat <- iris[, 1:4]

# Select a Gaussian mixture model using BIC
BIC <- mclustBIC(dat)
model2 <- Mclust(dat, x = BIC)

summary(model2)
```

```
## ----------------------------------------------------
## Gaussian finite mixture model fitted by EM algorithm
## ----------------------------------------------------
##
## Mclust VEV (ellipsoidal, equal shape) model with 2 components:
##
##  log-likelihood   n df       BIC       ICL
##        -215.726 150 26 -561.7285 -561.7289
##
## Clustering table:
##   1   2
##  50 100
```

```
# Visualize the estimated clusters
par(las = 1, mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 0.7, 0), family = "serif")
plot(model2, what = "classification", cex = 0.5)
```




```
# Compare estimated clusters with known species labels
table(Cluster = model2$classification, Species = iris$Species)
```

```
##         Species
## Cluster setosa versicolor virginica
##       1     50          0         0
##       2      0         50        50
```

The mixture model identifies groups using only the flower measurements. The species labels are used afterward to evaluate how well the estimated clusters correspond to the known botanical categories.

Keep in mind that the number of Gaussian mixture components need not equal the number of species. A single species may require multiple Gaussian components to represent its distribution.